

ONLINE RETAIL & E-COMMERCE DATA

1.PROJECT OBJECTIVES

This project focuses on analyzing an Online Retail & E-commerce dataset by performing data cleaning, transformation, and visualization. The raw dataset contains customer transactions, including product details, pricing, payment methods, and customer demographics. Initially, data preprocessing is carried out using Microsoft Excel and Power Query, where missing values are handled, inconsistent data is corrected, and the dataset is structured properly. After cleaning, the refined data is imported into Power BI to create interactive dashboards and visualizations. The dashboard provides insights into sales performance, customer behavior, product trends, and payment preferences, helping stakeholders make data-driven decisions.

2.Data Sources

The dataset used in this project represents an Online Retail & E-commerce transaction system. It contains detailed information about customer purchases, including product details, transaction values, payment methods, and customer demographics.

3.Problem Statement

- Track total revenue, total orders, total products sold, and average customer ratings.
- Identify which payment methods are used the most.
- Analyze product categories and find the best-selling categories and products.
- Monitor revenue trends over time to understand seasonal or monthly sales performance.
- Compare customer distribution by gender and city.
- Provide interactive filters for category, payment method, and city to support faster decision-making.

4.Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Customer ID	Integer / Text	Unique identifier for each customer
Order Date	Date	Date on which the order was placed
Product ID	Integer	Unique identifier for each product
Category ID	Integer	Unique identifier for product category
Category Name	Text	Name of the product category (e.g., Electronics, Fashion)
Product Name	Text	Name of the product purchased
Quantity	Integer	Number of units purchased
Price	Decimal	Price per unit of the product
Payment Method	Text	Mode of payment (Credit Card, Cash on Delivery, Bank Transfer)
City	Text	Customer’s city/location
Review Score	Integer	Customer rating (1 to 5 scale)
Gender	Text	Gender of the customer (M/F)
Age	Integer	Age of the customer

5.Tools & Technologies

- **Excel:** Used for initial data handling, including data cleaning, removing duplicates, formatting columns, and basic analysis using functions and Pivot Tables.
- **Power Query (Excel):** Applied for advanced data transformation such as handling missing values, filtering data, splitting columns, and standardizing formats before analysis.
- **Power BI:** Used for data modelling, creating relationships, applying DAX calculations (e.g., Total Sales, Average Rating), and building interactive dashboards with charts and slicers.

6.Data Pre-Processing (Excel / Power Query)

Tasks Performed:

1.Data Cleaning & Transformation:

Removed duplicate records, handled missing or inconsistent values, corrected data types (Date, Text, Number), and standardized formats (e.g., payment methods, category names). Created calculated fields such as Total Sales (Quantity × Price) for better analysis.

2.Filtering & Sorting:

Filtered irrelevant or incomplete records and sorted data based on key fields like Order Date, Category, and City to improve data organization and readability.

3.Pivot Tables (Excel):

Created Pivot Tables to summarize data and gain initial insights, such as:

- Total sales by category
- Sales by city
- Customer distribution by gender
- Average review scores

4.Data Transformation using Power Query:

Applied transformations like splitting columns, renaming headers, removing null values, and ensuring consistency across the dataset before loading into Power BI.

5.Fact and Dimension Table Creation (Data Modeling):

Structured the dataset into:

- Fact Table: Contains transactional data (Customer ID, Product ID, Quantity, Price, Total Sales, Order Date)
- Dimension Tables:
 - Customer Dimension (Customer ID, Gender, Age, City)
 - Product Dimension (Product ID, Product Name, Category Name)
 - Date Dimension (Order Date, Month, Year)

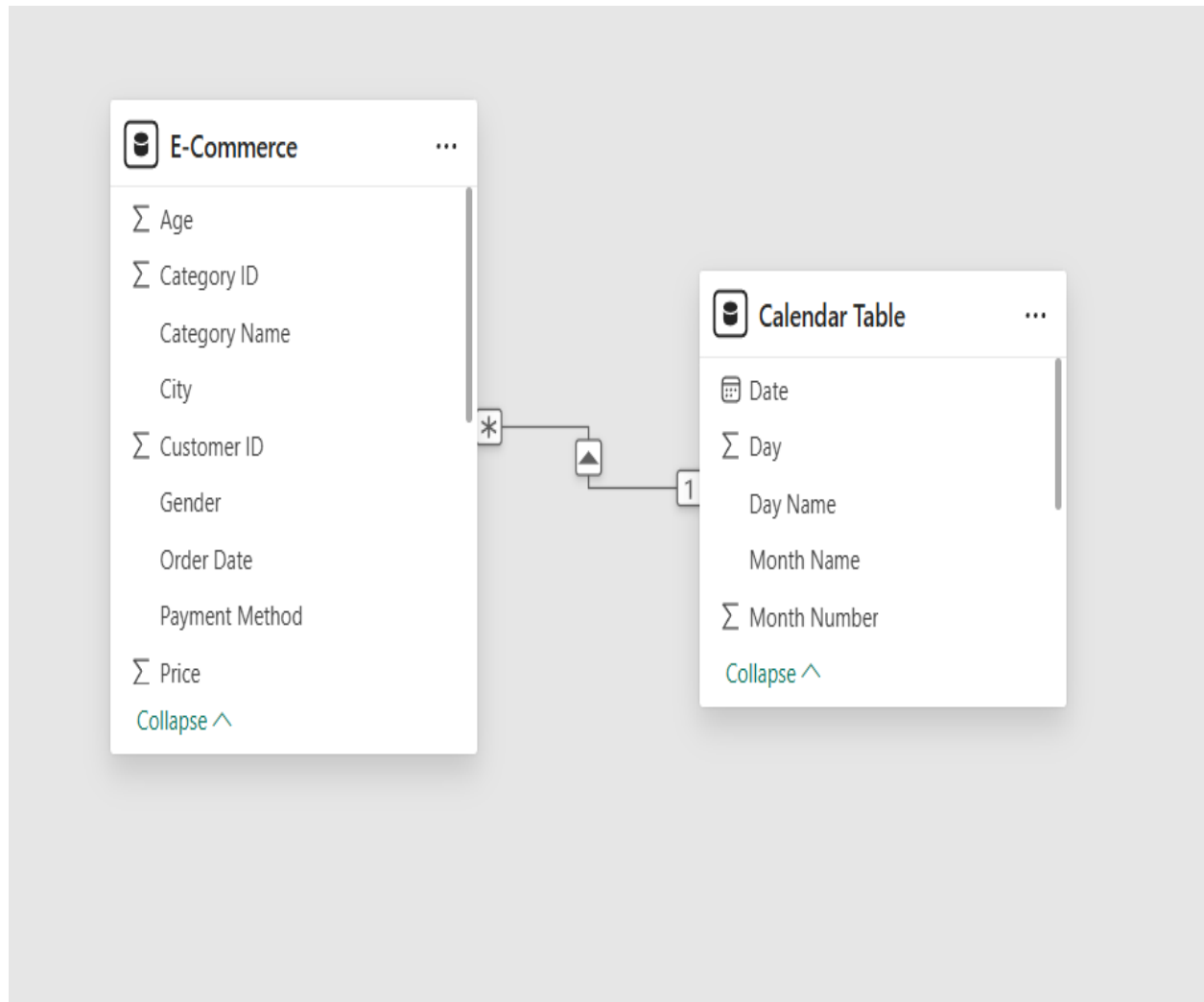
7. Data Modelling (Power BI)

Data Model :

The dataset was structured using a star schema model to improve performance and enable efficient analysis.

- Fact Table (E-Commerce):
Contains transactional data such as Customer ID, Product ID, Order Date, Quantity, Price, and Total Sales.

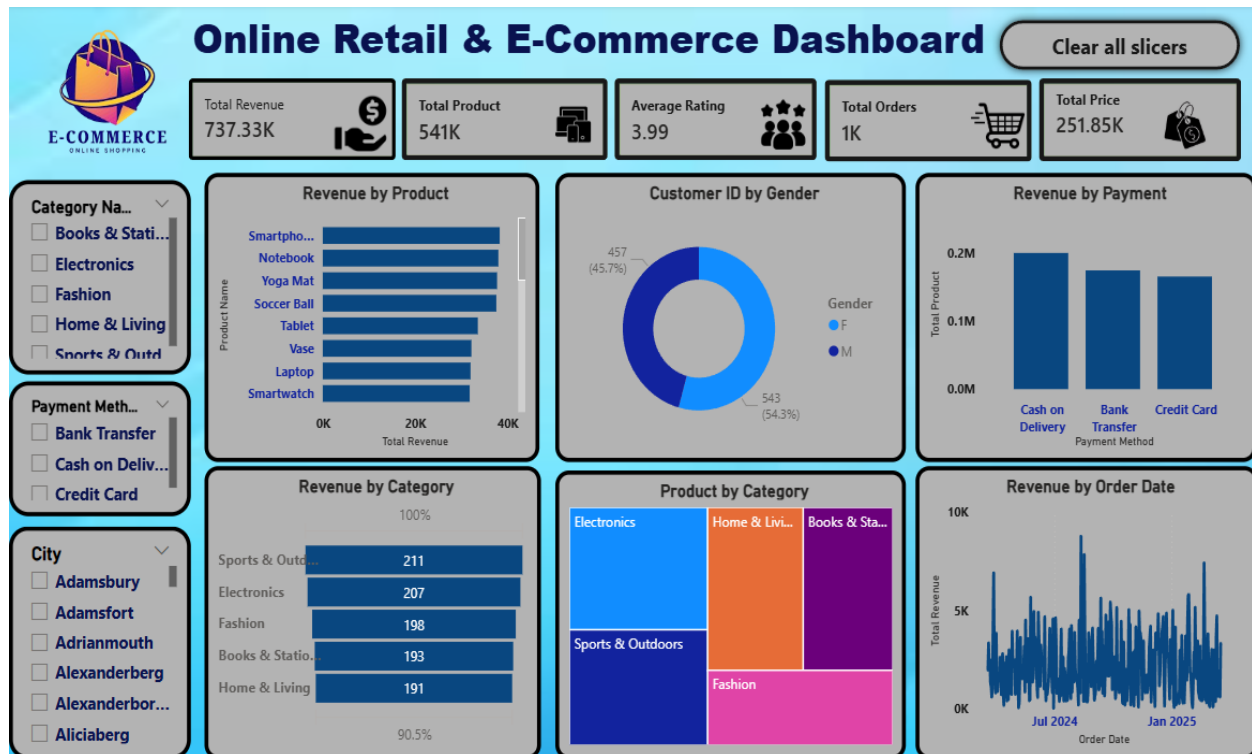
- Dimension Tables:
 - Date Table: Order Date, Month, Year
- Relationships Established:
 - Order Date → Date Table (Many-to-One)
- Cardinality:
 - Fact to Dimension: Many-to-One (*:1)
 - Cross filter direction: Single
- Lookup Tables Created:
 Separate dimension tables were created for Customer, Product, and Date to avoid redundancy and support better filtering.



8. Analysis and Visualizations (Power BI)

- Multiple Visualizations (Based on Problem Statement):
 The dashboard includes various charts to effectively present insights:
- Bar Chart:
 - 1.Revenue by Product
- Donut Chart:
 - 1.Customer by Gender
- Column Chart:

- 1.Revenue by Payment
- Funnel Chart:
 - 1.Revenue by Category
- Tree Chart
 - 1.Product by Category
- Line Chart
 - 1.Revenue by Order Date
- Cards (KPI Visuals):
 - 1.Total Revenue
 - 2.Total Product
 - 3.Total Orders
 - 4.Average Rating



9.Insights & Conclusions

- Revenue is highly concentrated among a small group of top-performing products, indicating a skewed sales distribution.
- Certain product categories consistently outperform others, contributing the majority of total revenue.
- Customer distribution by gender reveals a dominant segment, suggesting targeted purchasing behavior.
- A few payment methods generate most of the revenue, highlighting customer preference trends.
- Average review scores remain moderate, signaling opportunities to improve customer satisfaction.
- Some categories and products show low contribution despite availability, indicating underperformance or low demand.

1. Descriptive Analysis (What happened?)

- Total sales, quantity, and orders were analyzed across categories and time.
- Electronics and Home & Living categories generated significant revenue.
- Majority of transactions occurred through digital payment methods.

2. Diagnostic Analysis (Why did it happen?)

- High sales in Electronics are due to higher product prices and demand.
- Popular products like smartphones drive revenue growth.
- Higher ratings indicate good product quality and customer experience.
- Certain cities perform better due to higher customer activity or demand.

3. Predictive Analysis (What is likely to happen?)

- Demand for Electronics products is expected to continue growing.
- Customers will increasingly prefer **digital payment methods**.
- High-rated products are likely to maintain strong sales performance.
- Repeat purchases may increase among satisfied customers.

4. Prescriptive Analysis (What should be done?)

- Focus more on **high-performing categories** like Electronics to maximize revenue.
- Improve marketing strategies in **low-performing cities**.
- Encourage digital payments through offers and discounts.
- Maintain product quality to ensure high customer satisfaction.
- Introduce promotions for low-rated or low-selling products to boost sales.

10. Conclusions

The integration of Excel and Power BI proved highly effective for performing end-to-end data analysis, transforming raw e-commerce data into meaningful insights and interactive visual reports. Through data cleaning and preprocessing in Excel and Power Query, the dataset was structured and made analysis-ready. Power BI further enabled efficient data modeling, DAX-based calculations, and the creation of dynamic dashboards. Overall, this project demonstrates how combining Excel and Power BI can support data-driven decision-making, improve understanding of customer behavior, and enhance business performance through clear and actionable insights.